Diffusion Proposals for Redistricting Markov Chains

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Abstract

Recombination (ReCom) chains are the standard Markov chain Monte Carlo tool for sampling redistricting plans, and they are routinely run under a Metropolis filter that nominally targets a compactness-tempered distribution. We introduce a proposal built from graph diffusion. One convex energy on the merged pair of districts, a kernel-weighted Laplacian smoothness term plus a fidelity term that anchors each move to the current plan, is minimized under global population balance and rounded at random. The diffusion view yields a small theory of the proposal. Its optimum is a single linear solve whenever balance is slack, the expected number of units a proposal moves is an explicit high-pass filter norm with a provable ceiling, and mobility is governed by feasibility rather than by the Metropolis ratio, which gives a tuning rule no existing redistricting proposal has. Correcting the rounding with an exact Metropolis–Hastings step gives, to our knowledge, the first optimization-based redistricting proposal with a detailed-balance proof. Mixed with an exact tree move Metropolized to the same target, it yields a reversible sampler for kernel-weighted compactness distributions, validated against enumerated targets; the local kernel adds nothing on an exact-balance grid and is what reaches equilibrium on Iowa under perimeter and density targets. On the enumerated test bed we also show, at the level of distributions, that plain and tempered ReCom do not sample their nominal target and sit close to the spanning-tree-weighted measure. The same energy solved as an integer program is an anchored refinement tool, evaluated as an optimizer in a companion paper.

1 Introduction

1.1 The redistricting problem and ReCom's role

Political redistricting seeks a partition of a geographic region into electoral districts satisfying legal, demographic, and geometric requirements, namely contiguity, population balance, compactness, and preservation of communities of interest. Computational approaches model the problem on the dual adjacency graph $G = (V, E)$ of geographic units (precincts, census blocks, counties). A valid k-district plan is a partition

$$P = \{V_1, \dots, V_k\}, \qquad \bigsqcup_{i=1}^{k} V_i = V,$$

where each $G[V_i]$ is connected and the populations $p(V_i)$ are approximately equal. Let $\mathcal{P}_k(G)$ denote the space of valid k-district plans. The cardinality of $\mathcal{P}_k(G)$ is enormous. Even four congressional districts over the 99 counties of Iowa admit roughly 10^{24} plans [28].

Markov chain Monte Carlo (MCMC) has become the standard tool for sampling $\mathcal{P}_k(G)$. Early flip chains modify a single precinct's assignment per step [27] and mix slowly. The Recombination (ReCom) algorithm of DeFord, Duchin, and Solomon [22] performs multi-precinct updates by merging two adjacent

districts, sampling a uniform random spanning tree of the merged subgraph, and cutting an edge to recover a balanced bipartition. ReCom dramatically improved the mixing efficiency of redistricting MCMC and is now the workhorse of the field, with reversible [11] and multi-scale [3] extensions.

ReCom's blind spot. The uniform spanning-tree split treats every edge of the merged subgraph identically. The proposed bipartition is determined by the topology of the merged subgraph alone. Real geography is not topology-only. Shared boundary length, population density, demographic similarity, and administrative boundaries all carry signal that could shape the proposed split. Practitioners have felt this. County-weighted trees [19] and GerryChain's region-aware surcharges [41] inject indicator-based edge weights into the tree draw, and SpecReCom [21] replaces the tree with a Fiedler-vector cut of a randomly weighted Laplacian. But these mechanisms are restricted (region indicators, hard-coded county kernels, or no kernel at all), none includes a locality-preserving fidelity term, and, as their authors acknowledge, each shifts the chain's stationary distribution in an uncharacterized, uncorrected way.

1.2 Contribution

We introduce a diffusion proposal for ReCom-style MCMC and the exact sampler built on it. The core object is a single Laplacian energy on the merged-pair subgraph $H = G[V_i \cup V_j]$,

$$E(x) \;=\; \alpha\, x^\top L\, x \;+\; \beta\, \|x - x_0\|^2, \qquad (1-\varepsilon)\bar{p} \;\le\; p^\top x \;\le\; (1+\varepsilon)\bar{p}, \tag{1}$$

where L is the kernel-weighted graph Laplacian of H, x_0 encodes the current bipartition, and the balance constraint is anchored to the global per-district ideal (Equation (4)). The Laplacian term $x^\top Lx$ counts (or approximates) cut edges between the two sides; the fidelity term $\|x - x_0\|^2$ penalizes deviation from the current assignment, enforcing locality of each move. The paper makes three contributions, and a fourth item marks what is left to a companion paper.

1. **A diffusion proposal with a theory of its own behavior** (§4.2–§4.3). The interior optimum of the energy is one implicit Euler step of the graph heat equation with timestep $\tau = \alpha/\beta$. We show that this step is a single linear solve whose output lies in the unit cube automatically, so the box constraints never bind and the proposal costs a linear solve whenever population balance is slack (Lemma 4.1). Under single-shot Bernoulli rounding, the expected number of units a proposal moves is exactly the one-norm of a high-pass filter applied to the current split (Proposition 4.2). That quantity grows like twice the timestep times the pair's boundary for small timesteps and is bounded above by half the merged pair for any timestep (Corollary 4.1), so mobility cannot be bought by tuning. We verify each statement against the solver and measure the missing factor, acceptance, directly. Given a feasible proposal the Metropolis step accepts it at least 98 percent of the time; the chain's stickiness is entirely a feasibility effect with an interior optimum in τ (§4.6), and its mobility is given to within counting noise by a two-swap law with an explicit Metropolis factor, Equation (10). To our knowledge no other redistricting proposal comes with a rule for sizing its moves.

2. **An exactly reversible correction** (§4.5). Correcting the rounding with an exact Metropolis–Hastings step yields Chain E, to our knowledge the first optimization-based redistricting proposal with a detailed-balance proof (Proposition 4.3), stated on labeled plans, where the proposal map is a bijection. The proof identifies global anchoring of the balance window as the property that keeps forward and reverse solves symmetric. We verify the implemented kernel to machine precision on an enumerated fiber (Remark 4.1) and, on the same fiber, show by a batch-means test that a long run's occupancy distribution is not distinguishable from the exact target (§5.2). The ceiling of contribution 1 makes a global move necessary for mobility, so we pair Chain E with an exact

tree move Metropolized to the same target (§4.7), the reversible-recombination construction stated at the partition level, and verify the mixture to machine precision. On the enumerated fiber the mixture samples kernel-weighted targets correctly and the local kernel adds no mixing per second; on Iowa, where the tolerance makes single flips feasible and a uniform tree edge rarely balances fifty counties, matched two-million-step runs on two seeds give the hybrid 3.1 to 3.7 times the tree move's effective samples per second under the perimeter target and 4.7 to 6.7 under density, at the same equilibrium (§5.3).

3. **A ground-truth calibration of what the ReCom family samples, at the level of distributions** (§5.1–§5.1). On a fully enumerated 5×5 fiber we compute the spanning-tree-weighted measure for every state by the Matrix–Tree theorem and compare full empirical distributions against it and against the nominal tempered target, with sampling-noise floors at the effective sample size. The tempered chain's cut distribution is 220 floor standard deviations from its nominal target and close to, though distinguishable from, the tree-weighted tempered measure, while the exact chain is indistinguishable from the nominal target and far from the tree measure. The two chains separate exactly as theory predicts, which makes exact on enumerated ground truth what Clelland et al. [18] observed empirically.

4. **A companion optimizer.** The same energy solved as a convex mixed-integer program inside a tempered chain is an anchored, kernel-aware refinement tool in the matheuristic lineage [8, 12]. It is not a sampler, and its evaluation as an optimizer, against annealing baselines and in a least-change setting, is the subject of a companion paper [36]. A structural diagnostic of the partition fiber (Appendix B) supports the main contributions.

Positioning. The construction of Chain E, a deterministic optimization that sets the parameters of a random proposal which is then corrected exactly, has ancestors outside redistricting. Randomize-then-optimize [5] samples posteriors by perturbing and solving, and Zanella's informed proposals [57] give the general theory of locally balanced, target-aware proposals on discrete spaces. Chain E is an instance of that program on the space of plans, and our priority claim is scoped to redistricting accordingly. Within redistricting, the known-target samplers, Metropolized forest recombination [4], RevReCom, Cycle Walk, Marked Edge Walk, and BUD [11, 24, 39, 1], achieve computable transition probabilities by lifting to spanning-forest state spaces, while Chain E achieves the same property at the partition level; the two routes are complementary, and their global moves are the mobile component that our exact-but-local kernel provably lacks (Corollary 4.1). Bonsai sampling [20] draws independent plans from a specified law without a chain and is the natural non-MCMC comparison for any known-target method. §2 positions the proposal against the weighted-tree, spectral, and matheuristic lines.

Roadmap. §2 reviews related work and §3 sets up notation. §4 develops the proposal, its theory, and the exact chain; §5 reports the calibration, the convergence validation, the hybrid's validation and kill test, and the Iowa calibration. Appendices hold hyperparameters, the partition-fiber diagnostic, and a reproducibility checklist.

2 Related work

MCMC for redistricting. Early MCMC for redistricting framed the problem as a graph cut and used single-unit flip chains [27] or Swendsen–Wang-style large-swap moves. Flip chains mix poorly and produce highly irregular districts [22, 42]. ReCom [22] introduced the spanning-tree merge-split move and is now standard. Extensions include Multi-Scale Merge–Split [3] preserving nested communities (counties,

precincts), Forest ReCom [4], Sequential Monte Carlo [37] and its generalized form [43], whose repetition effects are analyzed in [10], multiscale parallel tempering [16], and reversible recombination [11] targeting the canonical spanning-tree distribution. The compactness-tempered target that practitioners run under a Metropolis filter, and that we audit here, originates in the North Carolina analyses of Herschlag et al. [32]. The 50-State Redistricting Atlas [38] is the field's standard ensemble benchmark. A consistent theme of the 2022–2026 literature is that a sampler is only as defensible as its target. RevReCom [11], Metropolized forest recombination [4], the cycle walk [24], and the marked edge walk [39] all pay a structural price (lifted state spaces, spanning-forest counting) precisely to keep transition probabilities computable. On the theory side, Cannon, Pegden, and Tucker-Foltz [13] give a polynomial-time sampler for the spanning-tree measure on balanced partitions of grids with a constant number of districts, Gold [30] exhibits planar-grid configurations from which recombination needs exponentially many expected steps to re-split, and mixing-time bounds for the local chains used in practice remain open. Akitaya et al. [2] prove that deciding reachability between balanced partitions under recombination is PSPACE-complete when the district count and step size grow with the graph.

Weighted and geography-aware proposals. Edge weighting inside the ReCom proposal is not new, and we position our work against it explicitly. Clelland et al. [19] replace the proposal's edge-weighting function with *county-weighted trees* (intra-county edge weights drawn from the uniform distribution $\mathcal{U}[0, w]$ for a tunable scalar w) to reduce county splits in Colorado ensembles. GerryChain's region-aware ReCom [41] implements the same idea. User-specified surcharges are added to the randomized Kruskal weights for edges crossing designated region boundaries. Tapp [50] modifies the ReCom tree-and-cut step to improve shape-based compactness at the proposal level. In all three, the weighting is an indicator- or attribute-indexed heuristic, and (as their authors acknowledge) the induced stationary distribution is uncharacterized and uncorrected. What the unweighted tree proposal itself induces is better understood. Clelland et al. [18] measured ReCom ensembles on grids and precinct graphs and found the sampling probability of a plan to decay roughly exponentially in its cut length, and Procaccia and Tucker-Foltz [45] proved the underlying fact that compact plans have many spanning trees. Tapp, Proebsting, and Ramsay [51] study how tolerance and county-preservation settings move ReCom ensembles and propose effective-sample-size diagnostics for them. Our calibration in §5.1 makes the first of these observations exact on an enumerated fiber, at the level of full distributions. Conversely, the samplers with known targets cited above handle only unweighted spanning-tree/forest measures or other tree-count-based families. The gap this paper addresses sits at the intersection, namely a proposal that accepts an *arbitrary* nonnegative edge kernel as a first-class input together with an exact accept/reject step against a stated target (§4.5, §4.7).

Optimization-based districting and matheuristics. A parallel operations research (OR) literature treats districting as exact optimization. Examples include the Hess model lineage, column-generation and set-partitioning formulations [31], multilevel multiobjective methods [49], contiguity-constrained MIPs [54], and cut-edge minimization solved to optimality at county/tract scale [53]. Dobbs et al. [25] solve for minimal-change and midpoint plans under transfer distance to reference plans exactly by mixed-integer programming, which is the natural exact benchmark for any anchored refinement. Between the sampling and optimization families [6], hybrids are emerging. Short bursts [12] use ReCom as a stochastic optimizer; Brous and Shmoys [8] replace the spanning-tree re-split with a Gurobi integer program inside an accept-if-better local search for majority-minority districts. The integer form of our energy, a convex MIQP refinement inside a recombination chain, belongs to this matheuristic line and is evaluated as an optimizer in the companion paper [36]. The proximal term $\beta\|x - x_0\|^2$ is the quadratic analogue of local branching [29], imposing a soft trust region around the incumbent.

Spectral partitioning and diffuse-interface models. Spectral graph partitioning has a long history in image segmentation and clustering [46]. In redistricting, the recent SpecReCom of Davies, Job, Kampbell, Kim, and Seo [21] (arXiv preprint, June 2025) replaces the random spanning tree with a Fiedler-vector sign cut on a randomly edge-weighted Laplacian of the merged pair, reporting roughly half the cut edges of RevReCom on a 56×56 grid and a Colorado planar graph. Plain SpecReCom sacrifices population balance, which their BalSpecReCom variant partially restores. Diffuse-interface models build on the Ginzburg–Landau (GL) functional, combining a smoothness term with a fidelity term; the Chan–Vese model [14] and Merriman–Bence–Osher (MBO) threshold dynamics implement this iteratively. Bertozzi and collaborators [7, 40] port the machinery to weighted graphs, where the smoothness term becomes the Laplacian quadratic form $x^\top Lx$ and graph Tikhonov regularization [44, 47] recovers heat-kernel smoothing. Jacobs and Walch [34] brought threshold dynamics to redistricting before SpecReCom, using volume-constrained MBO via auction dynamics [33] to generate compact equal-population maps. Theirs is a continuum PDE approach, not a graph-kernel one, and it is not embedded in MCMC. To our knowledge the present work is the first to use the *graph* GL lineage (kernel-weighted Laplacian smoothing plus fidelity) as a proposal inside redistricting MCMC.

Cut minimization and discrete compactness. The minimum graph cut [48, 35] and its balanced variants such as normalized cut [46] are classical objectives. NP-hardness of balanced minimum cut [55] motivates continuous relaxations, of which the Fiedler-vector cut and the Total Variation Isoperimetric Profile [23] are two well-known examples. The cut edge count $|\partial P|$ has been proposed as a discrete compactness measure for redistricting [26] and shown to correlate well with classical geometric scores (Polsby–Popper, Reock) while being more stable under unit refinement.

3 Background: plans and ReCom

3.1 Dual graph and k-district plans

Let $G = (V, E)$ be the dual graph of a state, with vertices representing geographic units (precincts, census blocks, counties) and edges encoding shared boundaries. Each vertex v carries a population $p_v \geq 0$ and may carry additional attributes (demographic vectors, shapefile geometry). A *k-district plan* is a partition $P = \{V_1, \ldots, V_k\}$ of V such that each induced subgraph $G[V_i]$ is connected (*contiguity*) and the populations satisfy

$$(1-\varepsilon)\bar{p} \;\leq\; p(V_i) \;\leq\; (1+\varepsilon)\bar{p}, \qquad \bar{p} := \frac{p(V)}{k}, \tag{2}$$

for a tolerance $\varepsilon \in [0, 1)$. We write $\mathcal{P}_k(G; \varepsilon)$ for the set of valid plans. The *cut edges* of a plan are

$$\partial P \;=\; \{(u, v) \in E : P(u) \neq P(v)\},$$

and we call $|\partial P|$ the *cut-edge count* of P. This quantity is the discrete analog of the perimeter of the district boundaries and serves as our primary compactness metric [26].

3.2 Recombination (ReCom)

Recombination [22] is a Markov chain on $\mathcal{P}_k(G; \varepsilon)$ whose single step is given in Algorithm 1.

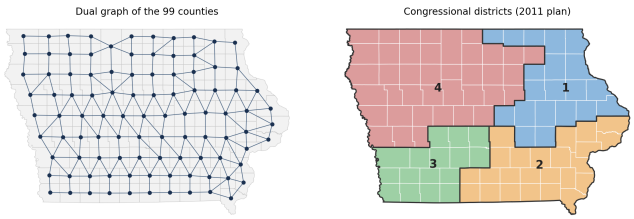


Figure 1: The dual graph of Iowa's 99 counties (left) and the four congressional districts enacted after the 2010 census (right). Nodes sit at county centroids; two counties are joined by an edge when they share a boundary segment.

Algorithm 1: Vanilla ReCom step (Chain V)

Input: Current plan P, dual graph G, balance tolerance ε.

Output: Next plan P'.

Select two adjacent districts V_i, V_j uniformly at random from boundary pairs;

Form the induced subgraph $H = G[V_i \cup V_j]$;

repeat

  Sample a uniform random spanning tree T of H via Wilson's algorithm [56];

  Search edges of T for a cut e^* whose removal produces two ε-balanced components.

until *a cut e^* exists yielding two ε-balanced components*;

Define V_i', V_j' as the two components of $T \setminus \{e^*\}$;

$P' \leftarrow P$ with V_i, V_j replaced by V_i', V_j';

return P'.

The tree-based proposal flips many precincts simultaneously between V_i and V_j while preserving contiguity automatically (any edge removal from a tree produces two connected components). This is the key advantage over single-precinct flip chains, which traverse the state space too slowly to be practically useful.

Chains compared. Table 1 lists the chains of this paper. The companion paper evaluates the same energy solved as an integer program, together with a continuous relaxation and a spectral mixture, as optimizers.

Chain	Method	Mechanism	Stationarity claim
V	Vanilla ReCom [22]	spanning-tree split, always accept	unknown target
A	Tempered ReCom	ReCom + MH on cuts (symmetric-q)	approximate
E	QP-MH (this work, §4.5)	QP + Bernoulli rounding + exact MH	**exact**
T	Exact tree move (§4.7)	uniform tree, uniform edge, tree-count MH	**exact**
H	Hybrid (this work, §4.7)	mixture of E and T, any kernel-weighted target	**exact**

Table 1: The chains compared in this paper. The last column is the chain's relationship to its stated target. Chains E, T and H carry exact Metropolis–Hastings corrections; Chain A relies on the symmetric-q approximation (§4.5) and Chain V has no stated target.

Tempered ReCom uses a Metropolis–Hastings filter targeting $\pi(P) \propto \exp(-\lambda|\partial P|)$ with the symmetric-q approximation (treating the forward and reverse proposal probabilities as equal); see Appendix A for the locked-in $\lambda = 0.10$. All chains enforce the global two-sided balance window of Equation (4) on every move; we note (§5) that the raw GerryChain `recom` proposal alone does not.

The proposal we replace. The tree-and-cut step is kernel-blind (§1.1), and the cut edge is chosen only after the tree is drawn, so the proposal cannot anticipate the shape of the split. The diffusion proposal of §4 replaces that step with a single convex optimization of a kernel-weighted Laplacian energy plus a locality-preserving fidelity term. The energy $x^\top L\, x = \sum_{(u,v)\in E} w_{uv}(x_u - x_v)^2$ equals the weighted cut count for binary x, so minimizing it directly minimizes the discrete perimeter in the chosen kernel.

4 The QP-diffusion model and its theory

4.1 Locality kernels

Choosing the edge weights w_{ij} is the most important modeling decision. The kernel determines what *smoothness* means and therefore controls how district boundaries move under the relaxation. We restrict all weights to geographic neighbors,

$$w_{ij} = 0 \quad \text{if } (v_i, v_j) \notin E, \tag{3}$$

ensuring that diffusion operates only through geographic adjacency and preventing long-range interactions that could produce disconnected districts.

Uniform kernel (k_{unif}). $w_{ij} = 1$ for every edge. The Laplacian L is the combinatorial graph Laplacian; $x^\top Lx$ is exactly the cut edge count. In the $\beta \to 0^+$ limit the proposal approaches a balanced Fiedler-style split.

Perimeter kernel (k_{perim}). $w_{ij} = |\partial v_i \cap \partial v_j|$ (shared-boundary length). The Laplacian quadratic form $x^\top Lx$ is then the *geographic perimeter* of the partition, weighted by boundary segment length. This kernel discourages fragmented districts and favors boundaries that cut along short segments.

Density kernel (k_{dens}). Let ρ_v denote the population density at vertex v (population per unit area). A Gaussian density kernel

$$w_{ij} \;=\; |\partial v_i \cap \partial v_j| \cdot \exp\Big(-\frac{(\rho_i - \rho_j)^2}{\sigma^2}\Big)$$

encourages cuts along sharp density gradients (urban–rural boundaries) while preserving the topological role of shared boundary length.

Demographic kernel (k_{demog}). Let $d_v \in \mathbb{R}^m$ be a vector of demographic attributes. A Gaussian similarity kernel $w_{ij} = \exp(-\|d_i - d_j\|^2/\sigma^2)$ encourages communities of interest. We define this kernel for completeness but do not use it in our experiments, and a caution stronger than "use as a soft regularizer" is warranted. If the attribute vector encodes race, a proposal that systematically shapes district boundaries around racial similarity risks making race the predominant districting criterion, which triggers strict scrutiny under the *Shaw v. Reno* line of racial-gerrymandering cases. Any practical use of demographic kernels should be reviewed against that doctrine, should document the kernel's weight relative to race-neutral criteria, and is more defensible for attributes that are legitimate districting factors in the jurisdiction (e.g., municipal membership, communities of interest as defined by state law).

Composite kernels. Linear combinations $w_{ij} = \sum_k a_k w_{ij}^{(k)}$ with $a_k \geq 0$ remain valid, since they preserve nonnegativity and locality.

4.2 The convex QP

Let $H = G[V_i \cup V_j]$ be the merged subgraph of two adjacent districts. Let L_H be the kernel-weighted Laplacian of H, let $x_0 \in \{0,1\}^{|V_H|}$ encode the current bipartition (1 for vertices in V_i, 0 for vertices in V_j), and let p be the population vector restricted to H. Fix weights $\alpha, \beta > 0$ and balance tolerance $\varepsilon \in (0,1)$. Crucially, the balance constraint is anchored to the *global* per-district ideal $\bar{p} = p(V)/k$ from Equation (2), not to $p(V_H)/2$. Both sides of the split must individually satisfy the plan-validity bound, which yields the two-sided window

$$\ell \;=\; \max\big((1-\varepsilon)\bar{p},\; p(V_H) - (1+\varepsilon)\bar{p}\big), \qquad u \;=\; \min\big((1+\varepsilon)\bar{p},\; p(V_H) - (1-\varepsilon)\bar{p}\big). \tag{4}$$

Anchoring to $p(V_H)/2$ instead (a tempting simplification, and one present in an earlier version of our own implementation) only bounds the local split. Because $p(V_H)$ itself drifts as the chain moves population between pairs, district populations can ratchet outside the global tolerance by a factor of up to $(1+\varepsilon)$ per move on heterogeneous graphs. Our experiments audit the global bound at every step (§5).

Definition 4.1 (Continuous QP-diffusion)**.** *The continuous QP-diffusion bipartition of H is*

$$x^\star \;=\; \arg\min_{x \in [0,1]^{|V_H|}} \; \alpha\, x^\top L_H\, x \;+\; \beta\, \|x - x_0\|_2^2 \tag{5}$$

$$\text{subject to } \; \ell \;\le\; p^\top x \;\le\; u, \tag{6}$$

with ℓ, u as in Equation (4).

The first term in (5) is the kernel-weighted Laplacian smoothness energy. For binary x it equals the weighted cut count $\sum_{(u,v)\in E(H)} w_{uv}\, \mathbf{1}[x_u \neq x_v]$, and the continuous relaxation $x \in [0,1]^{|V_H|}$ extends this to a smooth objective on the unit cube. The second term is the *fidelity term*, which penalizes deviation from the current bipartition and ensures that each proposed move is local. The balance constraint enforces ε-balance on the proposed split.

Proposition 4.1 (Convexity and well-posedness)**.** *The objective in (5) is strictly convex on $[0,1]^{|V_H|}$ (since $\beta > 0$ and L_H is positive semidefinite). The feasible region defined by (6) is a nonempty polytope whenever the current plan is valid (it then contains x_0). Hence the QP admits a unique minimizer $x^\star$, computable in polynomial time by any convex QP solver.*

Spectral interpretation. Projecting the interior optimality condition $(\alpha L_H + \beta I)x^\star = \beta x_0$ onto the eigenbasis $\{\phi_k\}$ of L_H with eigenvalues $0 = \lambda_1 \le \lambda_2 \le \cdots \le \lambda_n$ yields the closed-form frequency response

$$\hat{x}^\star_k \;=\; \frac{\beta}{\alpha\lambda_k + \beta}\, \hat{x}_{0,k}, \qquad \hat{y}_k := \phi_k^\top y. \tag{7}$$

The QP therefore acts as a low-pass spectral filter on the prior x_0, with cutoff controlled by the ratio α/β. Equivalently, with $\tau := \alpha/\beta$, the optimum satisfies $(x^\star - x_0)/\tau = -L_H x^\star$, the implicit Euler discretization of the graph heat equation $\dot{x} = -L_H x$. Inspecting the leading eigenvectors $\phi_2, \phi_3, \ldots$ provides a sanity check on kernel choice. If they look like noise rather than coherent geographic regions, the chosen kernel is probably not informative.

4.3 Theory of the diffusion proposal

The spectral identity is usually read as interpretation. Worked through, it predicts how the proposal behaves, and every prediction below is checked against the solver. Write $\tau := \alpha/\beta$ for the diffusion timestep, $L = L_H$ for the kernel-weighted Laplacian of the merged pair, $n = |V_H|$, and $\mathbf{1}_i \in \{0,1\}^n$ for the indicator of V_i, so that $x_0 = \mathbf{1}_i$.

Lemma 4.1 (The box constraints never bind)**.** *For any nonnegative kernel and any $\tau > 0$, the matrix $(I+\tau L)^{-1}$ is symmetric, entrywise nonnegative, and has unit row sums. Hence the unconstrained minimizer $x^\star = (I+\tau L)^{-1}x_0$ of (5) is a convex combination of the entries of x_0 and lies in $[0,1]^n$, and the QP of Definition 4.1 reduces to this single linear solve whenever the balance constraint (6) is slack at $x^\star$.*

Proof. $I + \tau L$ is symmetric with positive diagonal, nonpositive off-diagonal entries, and row sums equal to one, so it is strictly diagonally dominant, hence a nonsingular M-matrix, and its inverse is entrywise nonnegative. Since $L\mathbf{1} = 0$ we have $(I+\tau L)\mathbf{1} = \mathbf{1}$ and so $(I+\tau L)^{-1}\mathbf{1} = \mathbf{1}$. Each row of the inverse is therefore a probability vector and $x^\star_v$ is a weighted average of entries of $x_0 \in \{0,1\}^n$. The objective is strictly convex, so if the unconstrained minimizer satisfies the one remaining constraint it is the constrained minimizer. □

Proposition 4.2 (Expected move size)**.** *Under the single-shot Bernoulli rounding of §4.5, and whenever the balance constraint is slack, the expected number of units of H that a proposal reassigns is*

$$\mathbb{E}[\#\text{flips}] \;=\; \|x^\star - x_0\|_1 \;=\; 2\,\mathbf{1}_i^\top (I+\tau L)^{-1}\mathbf{1}_j \;=\; \frac{2\,|V_i|\,|V_j|}{n} \;-\; 2\sum_{k\geq 2}\frac{(\phi_k^\top \mathbf{1}_i)^2}{1+\tau\lambda_k}, \tag{8}$$

where (λ_k, ϕ_k) are the eigenpairs of L with $\lambda_1 = 0$.

Proof. A unit with $x_{0,v} = 0$ flips with probability $x^\star_v$ and one with $x_{0,v} = 1$ with probability $1 - x^\star_v$; both equal $|x^\star_v - x_{0,v}|$ because $x^\star_v \in [0,1]$ by Lemma 4.1, which gives the first equality. Write $P = (I+\tau L)^{-1}$. For $v \in V_i$, $1 - x^\star_v = \sum_{u\in V_j} P_{vu}$ by unit row sums, and for $v \in V_j$, $x^\star_v = \sum_{u\in V_i} P_{vu}$; summing and using symmetry of P gives $2\,\mathbf{1}_i^\top P\,\mathbf{1}_j$. Expanding $P = \sum_k \phi_k\phi_k^\top/(1+\tau\lambda_k)$ with $\phi_1 = \mathbf{1}/\sqrt{n}$ gives the $k=1$ term $|V_i||V_j|/n$, and since $\phi_k \perp \mathbf{1}$ for $k \geq 2$, $\phi_k^\top \mathbf{1}_j = -\phi_k^\top \mathbf{1}_i$, which gives the sum. Clipping at $\delta = 10^{-12}$ changes the expectation by at most δn. □

Corollary 4.1 (Small-timestep law, monotonicity, and ceiling)**.** *Let $|\partial_w(V_i, V_j)|$ be the boundary of the pair measured in the proposal's kernel. Then $\mathbb{E}[\#\text{flips}]$ is increasing in τ,*

$$\mathbb{E}[\#\text{flips}] \;=\; 2\tau\,|\partial_w(V_i,V_j)| + O(\tau^2) \quad (\tau \to 0), \qquad \mathbb{E}[\#\text{flips}] \;\leq\; \frac{2\,|V_i|\,|V_j|}{n} \quad \textit{for every } \tau,$$

with equality in the limit $\tau \to \infty$. For a balanced pair the ceiling is $n/2$, half the merged pair.

Proof. Each subtracted term in (8) is nonnegative and decreasing in τ, which gives monotonicity and the bound, and the terms vanish as $\tau \to \infty$. For the small-τ law, expand $(1+\tau\lambda_k)^{-1} = 1 - \tau\lambda_k + O(\tau^2)$; the τ^0 terms cancel because $\sum_{k\geq 2}(\phi_k^\top \mathbf{1}_i)^2 = |V_i| - |V_i|^2/n = |V_i||V_j|/n$, and the τ^1 term is $2\tau\sum_k \lambda_k(\phi_k^\top \mathbf{1}_i)^2 = 2\tau\,\mathbf{1}_i^\top L\,\mathbf{1}_i = 2\tau|\partial_w|$. □

Three consequences follow. The continuous proposal is far cheaper than "solve a QP" suggests, since it is one linear solve in the common case. The move size is a design dial with a known scale, so to propose moves of about m units one sets $\alpha/\beta \approx m/(2|\partial_w|)$ while the small-timestep law holds. And the dial has a hard stop. No choice of α and β can lift the expected move above half the merged pair, so mobility cannot be bought by tuning, which is the structural reason a global move is needed alongside the exact local chain (§4.6, §4.7). The kernel enters through $|\partial_w|$, so a perimeter kernel takes larger steps exactly where the geographic border is long.

Verification. Across 10 766 linear solves on the two test beds the unconstrained optimum never left the unit cube, as Lemma 4.1 requires. On 120 merged pairs with balance slack, Gurobi's constrained solution and the linear solve agree to 1.1×10^{-4}, the solver tolerance, so the implementation matches the theory. The median relative error of the small-timestep law against the exact filter norm is 2.5% at

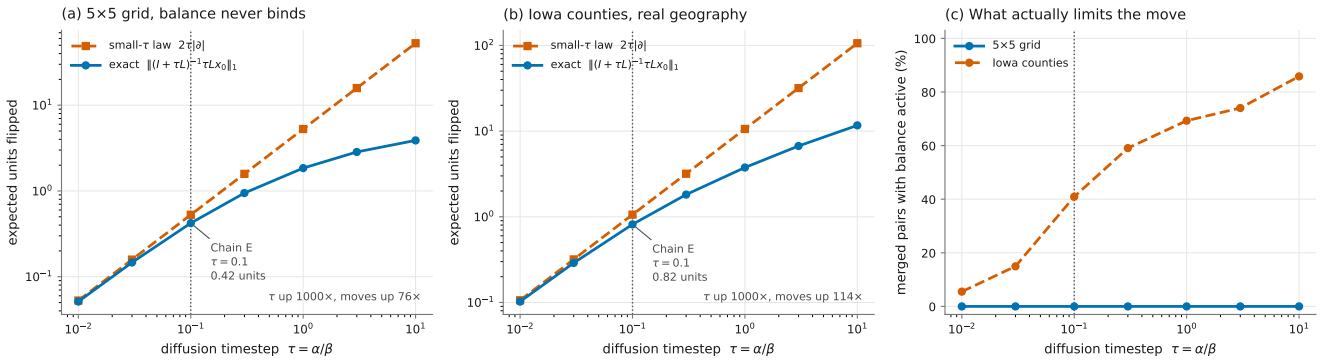


Figure 2: The move-size law against the solver. (a) On the enumerated 5×5 fiber, 1 411 merged pairs drawn from 200 plans; the exact filter norm of (8) and the small-timestep law $2\tau|\partial_w|$ agree for $\tau \leq 0.03$ and separate as the gain saturates. Raising τ by $1000\times$ raises the move by $76\times$, and the last decade buys $2.1\times$. The locked Chain E setting $\tau = 0.1$ is marked. (b) The same on Iowa, 127 pairs from 25 tree-partition plans. (c) Fraction of merged pairs whose balance constraint is active. On the uniform grid it never is, so the proposal is a linear solve at every step; on real county geography it is active for 41% of pairs at $\tau = 0.1$ and 86% at $\tau = 10$, so there the population constraint, not the diffusion, caps the move.

$\tau = 0.01$, 7.4% at 0.03, and 24% at 0.1 (Figure 2). At the paper's locked Chain E setting the expected move is 0.42 units on the 5×5 fiber and, on Iowa, 0.82 on the pairs whose window stays slack at every timestep and 1.26 averaged over all 127 pairs, about one unit per proposal, which is the quantitative content of the chain's stickiness reported in §5.1.

4.4 Deterministic rounding, and why it blocks the correction

The continuous optimum $x^\star \in [0,1]^{|V_H|}$ must be rounded to a bipartition. The natural deterministic choice is a Cheeger sweep [17], which sorts the entries of $x^\star$, scans the thresholds, keeps the balanced and connected candidate with the smallest cut, and repairs near-misses by flipping orphaned components to the other side. That is the rounding of the companion paper's continuous and integer refinement chains, and as a proposal it has a defect no repair fixes. An argmin-plus-sweep map is deterministic, so the reverse proposal has zero probability and the Hastings ratio degenerates; a chain built on it is a heuristic with no stationarity claim beyond ergodicity. §4.5 replaces the sweep with single-shot randomized rounding, which restores an everywhere-positive proposal density and an exact correction.

4.5 The exactly reversible QP-MH chain (Chain E)

Replacing the deterministic rounding of §4.4 with *single-shot randomized rounding* restores a tractable, everywhere-positive proposal density and yields an exactly reversible chain. One step of Chain E is as follows.

1. Pick an adjacent district pair (V_i, V_j) uniformly at random from the $|\mathrm{bp}(P)|$ boundary pairs of the current plan P;
2. Solve the (deterministic) QP of Definition 4.1 on H for $x^\star \in [0,1]^{|V_H|}$;
3. Draw the proposal $x' \in \{0,1\}^{|V_H|}$ by independent Bernoulli rounding from the *clipped* optimum, $x'_v \sim \mathrm{Bern}\big(\mathrm{clip}(x^\star_v; \delta, 1-\delta)\big)$ with $\delta = 10^{-12}$, *once*, with no retry on infeasibility. Clipping is part of the proposal's definition. The QP optimum can have active box bounds (coordinates exactly 0 or

1, e.g. at interior vertices pinned by the fidelity term), and sampling from the same clipped vector whose density enters the acceptance ratio is what makes the proposal law and its stated density identical, with every binary x' receiving positive probability;

4. Accept with probability

$$\min\Big(1,\ e^{-\lambda(|\partial P'|-|\partial P|)}\,\frac{q(x_0 \mid P', \{i,j\})}{q(x' \mid P, \{i,j\})}\,\frac{|\mathrm{bp}(P)|}{|\mathrm{bp}(P')|}\Big), \tag{9}$$

where $q(\cdot \mid P, \{i,j\})$ is the Bernoulli product density induced by the QP optimum at the corresponding state, and proposals that violate contiguity or the balance window (4) are rejected outright.

Proposition 4.3 (Detailed balance of Chain E)**.** *Chain E satisfies detailed balance with respect to* $\pi(P) \propto \exp(-\lambda|\partial P|)$ *restricted to* $\mathcal{P}_k(G;\varepsilon)$.

Throughout, fix the deterministic edge kernel w (weight jitter is never used in this chain) and write $\Omega = \mathcal{P}_k(G;\varepsilon)$ for the space of valid plans. Plans are *labeled* assignments in this subsection, so that a Bernoulli draw and its complement are distinct states and the proposal map is a bijection; on unlabeled partitions the map is two-to-one and the transition probability would need both terms. Since π depends only on $|\partial P|$ it is label-invariant, and forgetting labels after sampling recovers the unlabeled law. For a plan P and an adjacent pair $\{i,j\} \in \mathrm{bp}(P)$ write $H = V_i \cup V_j$ and let $x_0(P) \in \{0,1\}^{V_H}$ denote the indicator of V_i on H (the pair is drawn from a sorted list, so which district plays the role of i is a deterministic function of the unordered pair). Say that $P' \in \Omega$ is *pair-adjacent* to $P \in \Omega$ if $P' \neq P$, the two plans differ on exactly two districts $\{i,j\}$, and $H = V_i \cup V_j$ is connected. This relation is clearly symmetric in P and P'.

Lemma 4.2 (Proposal regularity)**.** *Let* $P, P' \in \Omega$ *be pair-adjacent via districts* $\{i,j\}$*, and suppose* $\beta > 0$*. Then:*

(i) $\{i,j\}$ *is the unique unordered pair whose selection can generate the move* $P \to P'$ *or the move* $P' \to P$*, and it is selectable from both endpoints:* $\{i,j\} \in \mathrm{bp}(P) \cap \mathrm{bp}(P')$.

(ii) The QP of Definition 4.1 on H *with the global window* (4) *has the same (nonempty) feasible set for the forward and reverse solves, and its optimum is unique, hence a deterministic function of* $(\textit{endpoint}, \{i,j\})$.

(iii) The clipped-Bernoulli density $q(\cdot \mid \cdot, \{i,j\})$ *is bounded below by* $\delta^{|V_H|} > 0$ *on every binary labeling of* V_H*; in particular* $q(x_0(P') \mid P, \{i,j\}) > 0$ *and* $q(x_0(P) \mid P', \{i,j\}) > 0$.

Proof. (i) A step that selects pair $\{a,b\}$ relabels vertices only between districts a and b and changes no other district. If $P' \neq P$, at least one vertex of H switched between i and j, so districts i and j both changed; any generating pair must therefore equal $\{i,j\}$. For selectability, note that (V_i, V_j) and (V_i', V_j') are both bipartitions of the connected graph H into nonempty parts (all four sets are nonempty districts of valid plans), and every such bipartition is crossed by at least one edge of H; hence the two districts are adjacent in both plans.

(ii) The feasible set is $\{x \in [0,1]^{V_H} : \ell \le p^\top x \le u\}$ with ℓ, u given by (4). It depends only on $(H, \varepsilon, \bar{p})$, not on the endpoint, so the forward and reverse solves optimize over the same set. It is nonempty at both endpoints because the window of (4) is precisely the condition that both sides of the split be valid districts, which $x_0(P)$ and $x_0(P')$ satisfy by validity of P and P'. This is where globality is essential. Anchoring the window to the merged-pair half $(1 \pm \varepsilon)\, p(V_H)/2$ instead lets a valid plan fail its own pair's window, zeroing a reverse density whose forward counterpart is positive and destroying reversibility outright. The

objective $\alpha\, x^\top L_H x + \beta\, \|x - x_0\|^2$ has Hessian $2(\alpha L_H + \beta I) \succ 0$ for $\beta > 0$, so the minimizer over the (convex, compact, nonempty) feasible set exists and is unique, and it is a function of (L_H, x_0, ℓ, u) only. Since the kernel w is deterministic, L_H depends only on H, and x_0 only on the endpoint and the pair.

(iii) Clipping maps the optimum into $[\delta, 1-\delta]^{V_H}$, so every Bernoulli factor of the product density lies in $[\delta, 1-\delta]$ and every outcome in $\{0,1\}^{V_H}$ has probability at least $\delta^{|V_H|}$. □

Proof of Proposition 4.3. Let K denote the transition kernel of Chain E and fix distinct $P, P' \in \Omega$. If P and P' are not pair-adjacent, no single step can produce either move. A step changes at most the two selected districts, so a generating pair would have to be the differing pair; if the plans differ on more than two districts no pair qualifies, and if the union H of the two differing districts is disconnected, that pair is not adjacent in either plan and is never selected. Hence $K(P, P') = K(P', P) = 0$ and detailed balance holds trivially.

Otherwise let $\{i, j\}$ be the unique generating pair, abbreviate $q_\to = q(x_0(P') \mid P, \{i, j\})$ and $q_\leftarrow = q(x_0(P) \mid P', \{i, j\})$, and note that both are positive and finite by Lemma 4.2(ii)–(iii). Because the rounding is single-shot, $q_\to$ is the exact probability that the proposal from P with pair $\{i, j\}$ equals P'. No renormalization by a feasibility event is involved (a retry-until-feasible loop would replace q by a conditional law whose state-dependent normalizing constant does not cancel in the ratio), and clipping *before* sampling makes the sampled law and the density in (9) the same object. Hence

$$K(P, P') \;=\; \frac{1}{|\mathrm{bp}(P)|}\, q_\to\, \alpha(P, P'), \qquad \alpha(P, P') = \min\Big(1,\; \frac{\pi(P')}{\pi(P)}\, \frac{q_\leftarrow}{q_\to}\, \frac{|\mathrm{bp}(P)|}{|\mathrm{bp}(P')|}\Big),$$

using $\pi(P')/\pi(P) = e^{-\lambda(|\partial P'| - |\partial P|)}$. Multiplying through,

$$\pi(P)\, K(P, P') \;=\; \min\Big(\frac{\pi(P)\, q_\to}{|\mathrm{bp}(P)|},\; \frac{\pi(P')\, q_\leftarrow}{|\mathrm{bp}(P')|}\Big),$$

an expression symmetric under exchanging $(P, q_\to)$ with $(P', q_\leftarrow)$. The identical computation started from P' gives the same right-hand side, so $\pi(P)K(P, P') = \pi(P')K(P', P)$.

It remains to check that the rejection pathways do not disturb this identity. A Bernoulli draw whose split leaves a side empty, disconnected, or outside the window (4) is a proposal onto a state of zero target mass and is rejected with probability one, contributing only to the diagonal $K(P, P)$, for which detailed balance is vacuous; the same applies to in-space proposals rejected by (9). Rejection on solver failure is symmetric because, by Lemma 4.2(ii), the solve is a deterministic function of $(H, \{i, j\}, \text{endpoint})$, so a failure that suppresses the move $P \to P'$ suppresses the reverse move identically. Detailed balance of K with respect to π on Ω follows, and with it stationarity of π [52]. □

Remark 4.1 (Scope, verification, and ergodicity)**.** *The proof treats the QP solve as exact; in floating point the minimizer is computed to solver tolerance, and determinism of the implementation (fixed kernel, no jitter, sorted pair enumeration) is what the argument actually uses. As an end-to-end check, we assembled the full transition kernel of the implementation on an exhaustively enumerated* 2×3 *grid* ($k = 2$, *tolerance* $\varepsilon = 1/3$, *so that the 18 labeled valid states include* 4+2 *as well as* 3+3 *splits) and verified* $\pi(P)K(P, P') = \pi(P')K(P', P)$ *entrywise, with maximum relative violation* 1.5×10^{-15} *(machine precision;* `db_check_2x3.py`*). Detailed balance gives stationarity, not ergodicity. Within a selected pair every binary relabeling of* H *has positive proposal probability by Lemma 4.2(iii), but irreducibility of Chain E across pairs on general* (G, k, ε) *is not established, so convergence statements are for the chain restricted to its communicating class. On the enumerated* 5×5 *fiber the chain's cut distribution is indistinguishable from its target (§5.1), on the* 2×3 *fiber a million-step run's full occupancy distribution is (§5.2), and the principled route to global mobility is the mixture of §4.7.*

Chain E trades mobility for exactness. Because a proposal must be feasible in a single Bernoulli draw, and because Corollary 4.1 caps how far a draw can reach, the chain behaves as a strictly local sampler. §4.6 measures exactly why, and the answer is feasibility rather than the Metropolis ratio. We therefore present Chain E as the local component of the sampler rather than as a practical standalone sampler. The principled route to mobility is precise. Mixing E with a global kernel preserves π only if that kernel is *also* π-invariant. RevReCom as published is reversible with respect to its spanning-tree measure, not our tempered π, so a naive E/RevReCom mixture would preserve neither; the correct construction Metropolizes the tree move toward π using its computable Hastings ratio [11, 4] and mixes *that* kernel with E, which is the construction of §4.7.

On the tempered baseline (Chain A). Our Tempered ReCom baseline applies the filter $\min(1, \exp(-\lambda\Delta|\partial P|))$ over the ReCom proposal, which *assumes a symmetric proposal*. ReCom's proposal is not symmetric; the true Hastings ratio involves spanning-tree counts of the affected districts [4, 11], so Chain A targets π only approximately, and §5.1 quantifies the bias on enumerated ground truth. We keep the symmetric-q baseline because it is what practitioners run.

4.6 Mobility of the exact chain: feasibility, not acceptance

Proposition 4.2 says how far a proposal reaches. Mobility is that reach times the probability the proposal is kept, and the second factor has two causes of different kinds. A Bernoulli draw can produce a split that is disconnected or outside the balance window, in which case it is rejected with probability one before any coin is flipped. Or it can be a valid plan that loses the Metropolis coin. We measured both on the enumerated 5×5 fiber, with $\alpha = \tau$ and $\beta = 1$, over 3 000 steps per timestep from the same initial plan (Table 2).

τ	$\mathbb{E}$[#flips]	single-flip share	feasible | single	Pr[feasible]	accept | feasible	units moved / step	states
0.01	0.036	0.035	0.000	0.964	1.000	0.0000	1
0.03	0.126	0.110	0.000	0.884	1.000	0.0033	6
0.10	0.385	0.268	0.000	0.683	0.999	0.0093	12
0.30	0.909	0.349	0.000	0.437	0.989	**0.0353**	**38**
1.00	1.914	0.254	0.000	0.193	0.983	0.0313	37
3.00	2.672	0.168	0.000	0.074	0.986	0.0187	19

Table 2: Chain E against the diffusion timestep on the 5×5 fiber ($k = 5$, exact balance). "Feasible | single" is the feasibility rate of single-unit proposals, "accept | feasible" the Metropolis acceptance rate among feasible proposals, and "units moved / step" the product of acceptance and accepted move size, the quantity a tuning rule should maximize. "States" counts distinct fiber states reached in 3 000 steps. The locked setting of Appendix A is $\tau = 0.10$.

Three facts follow. First, every single-unit proposal is infeasible at every timestep, exactly as exact balance requires, since reassigning one cell produces districts of sizes 4 and 6. The leading-order move of the diffusion proposal is therefore always rejected on this fiber, and the chain moves only through second-order coordinated draws. This is the mechanism behind the coverage figures of §5.1, derived rather than observed. Second, the Metropolis ratio is never the bottleneck. Given a feasible proposal, acceptance is between 98 and 100 percent throughout. The chain's stickiness is a feasibility effect, and the natural phrase "large moves are rarely accepted" is wrong in an instructive way. They are rarely feasible, and when feasible they are accepted. Third, acceptance rate is a misleading diagnostic for this chain. At $\tau = 0.01$ it is 96 percent while the chain never leaves its initial state, because 96 percent of proposals flip nothing. We report units moved per step instead.

Units moved per step has an interior maximum near $\tau = 0.3$. The setting locked in Appendix A, $\tau = 0.1$, sits about 3.8× below it, so on this fiber the chain as run in §5 is mistuned by its own theory and can be improved for free; §5.3 shows the optimum does not transfer to Iowa, where the balance window binds more often at larger τ. Even at the optimum a thousand steps move about 35 units, and Corollary 4.1 says no timestep can lift the per-proposal reach above half the merged pair. Both statements matter. Chain E can be several times more mobile than shown, and several times is not enough, which is why a global move is required (§4.7). Under a loose tolerance single flips become feasible and mobility gains a first-order term in τ, which is why the same chain climbs steadily on Iowa (§5.4) while freezing on the exact-balance grid.

The mobility law. On an exact-balance fiber the cheapest feasible draw is a swap, one unit $u \in V_i$ and one unit $v \in V_j$ changing sides together, which preserves both populations. Rounding flips units independently with probabilities $p_v = |x^\star_v - x_{0,v}|$, so the probability that exactly $\{u, v\}$ flips is $p_u p_v \prod_{w \neq u,v}(1 - p_w)$, and the two-swap channel contributes

$$M_2 \;=\; 2 \sum_{\substack{u \in V_i,\ v \in V_j \\ \text{swap feasible}}} p_u\, p_v \prod_{w \neq u,v} (1 - p_w)\; \alpha(u, v) \tag{10}$$

units per step, where the sum runs over cross-boundary pairs whose swap keeps both districts connected and $\alpha(u, v)$ is the Metropolis probability of that swap, each of which costs one reverse linear solve. For small τ, $p_v = \tau b_v + O(\tau^2)$ with b_v the kernel weight of v's cross-boundary edges, so $M_2 = 2\tau^2 S_2 + O(\tau^3)$ with $S_2 = \sum_{\text{feasible swaps}} b_u b_v$, a coefficient computable from the current plan. We checked the law by running Chain E for 20 000 steps per timestep and evaluating (10) at every step for the pair chosen and the plan visited (Table 3).

τ	measured units / step	M_2	M_2 without α	$2\tau^2 S_2$
0.03	0.0021	0.0018	0.0024	0.0033
0.10	0.0105	0.0105	0.0127	0.0341
0.30	0.0320	0.0326	0.0383	0.3628

Table 3: The mobility law against Chain E on the 5 × 5 fiber, 20 000 steps per timestep. The two-swap channel with its Metropolis factor matches the measured mobility to within counting noise at every timestep; the $\tau = 0.03$ run holds 21 accepted swaps.

The two-swap channel accounts for the chain's mobility at every timestep tested, and four-unit draws contribute nothing measurable even at $\tau = 0.3$. The Metropolis factor is not optional. Without it the channel overpredicts by about 20%, because the chain's near-unit acceptance given feasibility is dominated by zero-flip proposals, and a genuine swap that lengthens the boundary is accepted less often. The small-timestep form is accurate only for $\tau \lesssim 0.03$, a narrower range than the move-size law's, as expected for a second-order quantity. Under a loose tolerance single flips become feasible, the law gains a first-order channel, and the same calculation applies with single units in place of swaps.

4.7 An exact tree move, and the reversible hybrid

Corollary 4.1 says the local chain cannot be made mobile by tuning, so a global move is needed, and §4.5 says that move must be reversible for the same target. The tree move below is the standard construction of reversible recombination [11, 4], stated at the partition level and Metropolized to an arbitrary target. We include it because the paper needs a mobile exact kernel to pair with Chain E, and because its density is closed-form only under one specific choice that practice does not make.

One step. Pick an adjacent pair $\{i, j\}$ uniformly from $\mathrm{bp}(P)$, draw a uniform spanning tree T of $H = G[V_i \cup V_j]$ by Wilson's algorithm, and pick one edge of T uniformly among *all* $|V_H| - 1$ tree edges. If the two components A, B both lie in the window (4), assign the labels i, j to them by a fair coin and propose; otherwise reject. Every spanning tree of H that yields $\{A, B\}$ on removal of one edge is a spanning tree of $G[A]$, a spanning tree of $G[B]$, and one crossing edge, so

$$q(P \to P') \;=\; \frac{\tau(A)\,\tau(B)\,|\partial(A,B)|}{2\,|\mathrm{bp}(P)|\,\tau(H)\,(|V_H|-1)},$$

where $\tau(\cdot)$ is the spanning-tree count by the Matrix–Tree theorem and $|\partial(A, B)|$ the number of edges between A and B. The reverse move shares H, so $\tau(H)$ and $|V_H| - 1$ cancel and the move is accepted with probability

$$\min\Big(1,\; \frac{\pi(P')}{\pi(P)}\;\frac{\tau(V_i)\,\tau(V_j)\,|\partial(V_i,V_j)|}{\tau(A)\,\tau(B)\,|\partial(A,B)|}\;\frac{|\mathrm{bp}(P)|}{|\mathrm{bp}(P')|}\Big),$$

computable from four determinants and two edge counts. Two details carry the exactness. The fair coin makes the move a bijection on labeled plans; a deterministic labeling rule would leave some reverse moves with zero density. And the cut edge is drawn among all tree edges rather than among balanced ones. GerryChain's `recom` draws among balanced edges, which multiplies the density by a tree-dependent normalization that does not cancel, and that is precisely why Chain A is only approximately correct. The price of the closed form is acceptance. Most tree edges split a merged pair unevenly, so the move accepts about 7% of proposals on the exact-balance 5×5 fiber and under 2% on Iowa at $\varepsilon = 0.05$, and it is usable only because a step costs tens of microseconds.

The hybrid. Take a local step of §4.5 with probability p_{loc} and a tree step otherwise. Both kernels are π-reversible, so the mixture is. The target may be kernel-weighted, $\pi_w(P) \propto \exp(-\lambda\, w(\partial P))$, at the cost of replacing $|\partial P|$ by $w(\partial P)$ in both acceptance ratios; the local move's Laplacian then uses the same kernel, so the proposal is matched to the target's geometry. Where the balance window is slack the local solve is the linear solve of Lemma 4.1, which we use in place of the QP solver; both paths are deterministic functions of the state and pair, which is all Proposition 4.3 requires.

Proposition 4.4 (Ergodicity of the hybrid)**.** *If the tree move is irreducible on Ω (which holds under the standard assumptions for recombination moves; see [22, 9]), then the mixture with the local move at any $p_{\mathrm{loc}} \in (0, 1)$ is ergodic.*

Proof. Write $K = (1 - p_{\mathrm{loc}})K_{\mathrm{T}} + p_{\mathrm{loc}}K_{\mathrm{E}}$. If $K_{\mathrm{T}}^t(P, P') > 0$ for some t, then $K^t(P, P') \geq (1 - p_{\mathrm{loc}})^t\, K_{\mathrm{T}}^t(P, P') > 0$, so irreducibility of the tree move implies irreducibility of the mixture. The tree move rejects with positive probability at every state (any unbalanced edge draw), so $K(P, P) > 0$ and the chain is aperiodic. □

Verification. On the 18 labeled states of the 2×3 fiber we assembled the tree kernel, the local kernel with the linear-solve path, and their mixture analytically and checked $\pi(P)K(P, P') = \pi(P')K(P', P)$ entrywise. The maximum relative violation is 4.8×10^{-16} for the tree move, 1.6×10^{-15} for the local move, and 7.2×10^{-16} for the mixture, under both the uniform kernel at $\lambda = 0.1$ and a density kernel at $\lambda = 1$ (`hybrid.py`). The Matrix–Tree code returns $\tau(H) = 15$ for the 2×3 grid, the known value. §5.3 tests both kernels on the 4 006-state fiber.

5 Experiments

We evaluate the chains of Table 1 on the 5×5 grid (full enumeration, ground-truth target), the 2×3 grid (18 states, full convergence checkable), and the Iowa county dual graph (real geography, 99 counties, $k = 4$, $\varepsilon = 0.05$). All hyperparameters are locked in Appendix A; raw traces, per-step balance audits, and configuration stamps are released in the project repository.

Statistical protocol. Chains are autocorrelated, so we do not report i.i.d. test statistics on raw traces. Effective sample sizes (ESS) come from Geyer's initial-positive-sequence estimator of the integrated autocorrelation time, computed by FFT, and are appropriate for inference about cut means only; distributional tests use batch means (§5.2), and distances to enumerated targets are read against parametric-bootstrap noise floors at the ESS (§5.1). Each chain runs on its own seeded RNG stream; the baseline tables report seed 42, and the ten-seed means of Chains V and A (seeds 42 through 51) are quoted where they enter the calibration. Between-seed standard deviations of the baselines are about 0.4 cuts on Iowa and 0.1 to 0.2 cuts on the 5×5 fiber, so single-seed means are representative.

Balance audit. Every run of the baseline chains records, at every step, the maximum relative deviation of any district population from the global ideal, and all of them satisfy the $\varepsilon = 0.05$ bound at every step (Iowa maxima 0.0498–0.0500 for the runs of §5.4); the exact chains of §4.7 enforce the window inside each move by construction, rejecting any draw outside it. We flag this because it is not automatic. GerryChain's raw `recom` proposal validates only one side of each tree cut, and an earlier version of our own local steps anchored balance to the merged-pair half rather than the global ideal (§4.2); either slip lets district populations drift to $\sim 4\times$ the nominal tolerance on Iowa before the audit catches it. We recommend per-step balance auditing as standard practice for any custom chain harness that bypasses GerryChain's validator pipeline.

5.1 Ground truth on the enumerated fiber: what the family samples

The 5×5 partition fiber (the set of all valid plans; $k = 5$, exact balance, 4 006 partitions; Appendix B) admits full enumeration, so every distributional quantity is exactly computable. The chains run with $\varepsilon = 0.05$, which on 25 unit-population cells admits only districts of exactly five cells, so the state space the chains traverse is precisely this enumerated fiber. Uniform-on-fiber is not the target of any chain here; the total-variation column is a descriptive concentration measure only, and with about 900 correlated draws over 4 006 states even an independent uniform sampler would show TV above 0.78. The informative comparisons are the cut means against the exact target mean and the long runs of §5.1, where distances are read against proper noise floors.

Chain	Mean cuts	Std	Unique / 4006	TV(emp, uniform)
V: Vanilla ReCom	17.06	1.08	337 / 4006 (8.4%)	0.92
A: Tempered ReCom	17.02	1.15	311 / 4006 (7.8%)	0.92
E: QP-MH exact	16.27	0.44	6 / 4006 (0.1%)	1.00

Table 4: 5×5 short-run comparison ($k = 5$, exact balance, 1 000 steps, burn-in 100, seed 42). The exact target mean is $\mathbb{E}_\pi[|\partial P|] = 18.56$ at $\lambda = 0.10$, and *every* chain sits far below it at this horizon, including the exactly reversible Chain E, which at 1 000 steps has visited only 6 states. Exactness of the kernel does not buy mixing speed.

The origin of the gap. The gap between every chain's empirical mean and the target mean 18.56 quantifies the compactness bias *shared* by the whole ReCom-derived family, and on the enumerated fiber we can identify its origin. Using the Matrix–Tree theorem we compute, for each of the 4 006 partitions, the spanning-tree-count weight $\prod_i \tau(V_i)$, the measure that the RevReCom analysis identifies as the natural target of tree-based recombination [11, 4] and whose relation to compactness Procaccia and Tucker-Foltz [45] proved. At the level of means the match is close. The tree-weighted measure has $\mathbb{E}[|\partial P|] = 16.88$ against Vanilla ReCom's ten-seed mean of 16.95 ± 0.11, and the tree-weighted-and-tempered measure $\prod_i \tau(V_i)\, e^{-\lambda|\partial P|}$ has $\mathbb{E}[|\partial P|] = 16.78$ against Chain A's ten-seed mean of 16.71 ± 0.16;

both gaps are 0.07 cuts, within one between-seed standard deviation, and the two quasi-targets sit 1.7 and 1.8 cuts below the nominal tempered mean. A mean is a one-dimensional summary, though, and many distributions share one. The long runs below compare full distributions, which is where the claim is actually tested.

Long-run validation at the level of distributions. We run Chain A for 2×10^5 steps and Chain E for 10^5 steps on the enumerated fiber (seed 42, 10% burn-in) and compare each chain's empirical distribution against three exactly computable reference measures, the nominal target $\pi \propto e^{-\lambda|\partial P|}$, the tree-weighted measure $\prod_i \tau(V_i)$, and the tree-weighted-and-tempered measure. Two design choices make the comparison informative. First, a raw total-variation distance is uninterpretable on 4 006 states, since even an independent sampler shows large TV until the draw count greatly exceeds the state count. So every distance is reported against a parametric-bootstrap noise floor, the same distance computed for independent draws from the reference measure at the chain's effective sample size. A converged chain sits at its floor; a chain converging elsewhere sits above it by many floor standard deviations. Second, the headline statistic is the cut-count marginal rather than the full state distribution. The fiber's cut counts take five values, so with 1.8×10^5 retained draws the marginal is estimated to three decimals and its floor is small, which removes the coverage problem entirely.

Chain	Reference measure	TV	Floor at ESS	z
A (2×10^5 steps, ESS 12 312)	nominal π	0.629	0.007	+220
	tree-weighted	0.065	0.006	+22
	tree-weighted, tempered	0.022	0.006	+6.7
E (10^5 steps, ESS 28)	nominal π	0.079	0.141	−1.2
	tree-weighted	0.493	0.125	+6.9
	tree-weighted, tempered	0.532	0.125	+7.5

Table 5: Total variation between each chain's empirical cut-count marginal and three reference measures on the enumerated 5×5 fiber, with the parametric-bootstrap noise floor at the chain's effective sample size and the excess in floor standard deviations. Chain A is far from its nominal target and close to, though distinguishable from, the tree-weighted tempered measure. Chain E is indistinguishable from the nominal target it is proved reversible for and far from the tree measures.

Chain A does not sample its nominal target, and the gap is a plateau, not a drift. At 2×10^5 steps the tempered chain has visited 3 524/4 006 states, its mean is 16.73, and its cut distribution sits 220 floor standard deviations from the nominal target, at TV 0.63. Figure 3(b) shows that distance flat across five decades of run length while the distance to the tree-weighted tempered measure falls toward the floor. This is convergence to a different distribution, not slow mixing, and it is what the symmetric-q filter of §4.5 predicts. The match to the tree-weighted tempered measure is close but not exact. At TV 0.022 the chain sits 6.7 floor standard deviations above it, so the tree measure is a very good description of what tempered ReCom samples and not a perfect one, which is expected, since Vanilla ReCom's stationary law is not exactly the tree measure and RevReCom exists to make it so. On the full 4 006-state distribution the noise floor at effective sample size is 0.15 to 0.23, so that comparison can separate the nominal target ($z = +164$) but not the two tree measures from each other, which is why the marginal carries the claim.

Chain E samples what it states. At 10^5 steps its cut marginal is 1.2 floor standard deviations from the nominal target and about 7 from either tree measure. The two chains separate exactly as theory predicts. Convergence of the full distribution is far slower for this sticky local chain, as §4.6 explains. It has visited 403/4 006 states, and its full-state TV of 0.90 sits at a floor of 0.99 at effective sample size 28, so that test has no power here. The fiber on which full convergence can be checked is the next subsection.

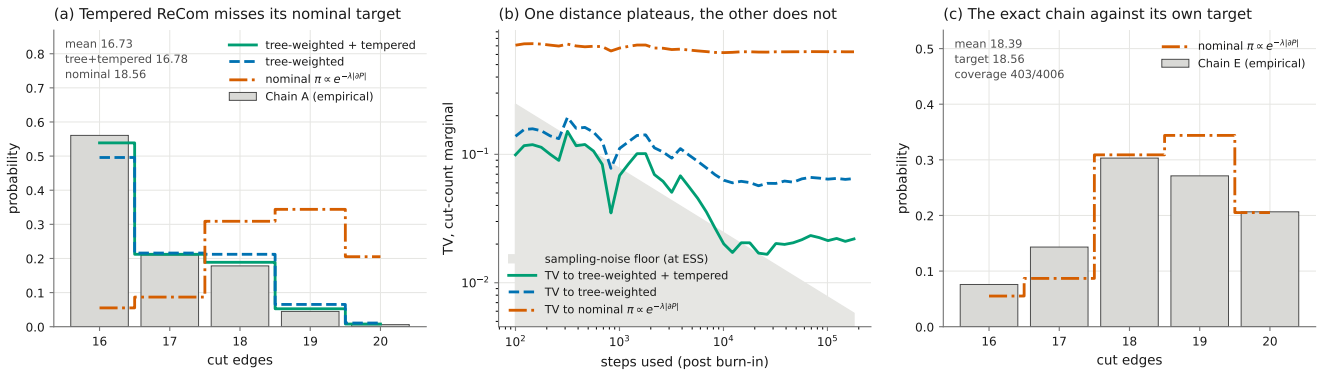


Figure 3: What the chains sample. (a) Chain A's cut-count marginal after 2×10^5 steps against the three reference measures. (b) Total variation of the running marginal to each reference as a function of run length, with the noise floor at the effective sample size shaded; the distance to the nominal target is a plateau across five decades while the distance to the tree-weighted tempered measure descends toward the floor. (c) Chain E's marginal after 10^5 steps against its own target.

5.2 Exact convergence on the 18-state fiber

Detailed balance gives stationarity, not convergence, and on 4 006 states the exact chain cannot be run to convergence in reasonable time. On the 2×3 grid with $k = 2$ and tolerance $\varepsilon = 1/3$, the fiber of labeled valid plans has 18 states, the same state space on which Remark 4.1 verifies the kernel to machine precision, and a long run can cover it. We run Chain E for 10^6 steps (seed 42, 10% burn-in, locked parameters $\alpha = 1$, $\beta = 10$, $\lambda = 0.10$). It visits all 18 states, its acceptance rate is 0.785, and its post-burn-in mean cut count is 2.75992 against the exact 2.76002.

The distributional test uses batch means, because thinning a chain by an effective sample size computed from a summary statistic is not valid here. The cut count takes two values on this fiber, so its integrated autocorrelation time (17 steps) understates that of the 18-state occupancy process (25 steps), and a chi-square test on a chain thinned by the former rejects spuriously. We therefore split the trace into 200 blocks of 4 500 steps, over 180 occupancy correlation times each, estimate the covariance of the occupancy vector from the between-block scatter, and test with Hotelling's T^2 on the 17 free coordinates. The result is $F(17, 183) = 0.87$, $p = 0.61$. The occupancy distribution is not distinguishable from the exact target. Total variation to the target is 0.0110, against a noise floor of about 0.009 at the occupancy effective sample size, and it fell from 0.0250 at 2×10^5 steps, a factor of 2.27 for five times the steps, which is the $\sqrt{n}$ rate of a converged chain. At 2×10^5 steps the same test gave $p = 0.07$ with three states in one cut-level set persistently under-visited; at 10^6 steps those deviations dissolved into the few $|z| \approx 2$ values that 18 states produce by chance. Together with the kernel check, this is end-to-end validation of the construction of §4.5. The implemented kernel satisfies detailed balance exactly, and the chain it generates reaches its target.

5.3 The exact tree move and the hybrid against enumerated targets

We run the tree move alone and the hybrid of §4.7 from the same start for 2×10^5 steps on the 5×5 fiber against three enumerated targets, the uniform kernel at $\lambda = 0.1$ and at $\lambda = 1$, and a density kernel at $\lambda = 1$ built from a synthetic central city, $\rho(x, y) = 3\exp(-r^2/2)$ with $w_{uv} = \exp(-(\rho_u - \rho_v)^2)$, under which π_w favors boundaries that follow the density gradient and the tree proposal is blind to the kernel. Each chain is tested as in §5.1, by the cut-count marginal against its noise floor and by batch-means Hotelling, and we report effective sample size per second of the weighted-cut trace. That ratio, hybrid to

tree-only, is the criterion we fixed before running. At or above 1.5 the local kernel earns its place, and below 1 it is a passenger. The hybrid was run at the paper's locked timestep with $p_{\text{loc}} = 0.5$, and at the mobility optimum of §4.6, $\tau = 0.3$, with $p_{\text{loc}} = 0.2$.

Target	Chain	TV marg	floor	z	Hotel. p	acc tree	acc loc	ESS	ESS/s	ratio
uniform, λ=0.1	tree only	0.010	0.039	−1.8	0.58	0.066	—	384	53.1	—
	hybrid τ=0.1, p=0.5	0.030	0.055	−1.2	0.27	0.064	0.657	188	10.5	0.20
	hybrid τ=0.3, p=0.2	0.028	0.039	−0.7	0.51	0.065	0.407	367	31.7	0.63
uniform, λ=1	tree only	0.028	0.040	−0.7	0.21	0.074	—	330	45.5	—
	hybrid τ=0.1, p=0.5	0.018	0.048	−1.6	0.88	0.074	0.674	231	12.3	0.27
	hybrid τ=0.3, p=0.2	0.013	0.033	−1.5	0.73	0.073	0.413	**534**	46.4	**1.07**
density, λ=1	tree only	0.028	0.039	−0.7	0.60	0.069	—	418	55.9	—
	hybrid τ=0.1, p=0.5	0.040	0.064	−1.0	0.50	0.067	0.785	136	6.4	0.11
	hybrid τ=0.3, p=0.2	0.033	0.045	−0.7	0.88	0.068	0.552	283	21.2	0.39

Table 6: The exact tree move and the hybrid on the enumerated 5×5 fiber, 2×10^5 steps each, seed 42. "TV marg" is total variation of the cut-count marginal to the enumerated target with its noise floor at the effective sample size and the excess z in floor standard deviations; "Hotel. p" is the batch-means test of §5.2; "acc" are acceptance rates by move type; ESS is of the weighted-cut trace and "ratio" is ESS per second relative to the tree move alone, timed in the same session, whose wall time in the $\tau = 0.3$ session differs from the displayed tree-only row by a few percent.

Three findings. First, every chain samples its target. All nine chains sit within 1.8 floor standard deviations of the enumerated marginal with Hotelling p between 0.21 and 0.88, under the uniform and the density kernel alike, so the exact tree move and the mixture are correct for kernel-weighted targets on 4 006 states and not only on the 18 states of the analytic check. Second, at the paper's locked timestep the local move is a passenger. With $p_{\text{loc}} = 0.5$ the hybrid's effective sample size per step is what its tree moves alone would give, and per second it is four to nine times worse, because a local step costs several tree steps and moves about 0.01 units. Third, retuning the local move to the mobility optimum changes the per-step picture but not the verdict. At $\tau = 0.3$ and $p_{\text{loc}} = 0.2$ the hybrid at the sharp uniform target reaches 534 effective samples against 330 for the tree move alone, with a fifth fewer tree moves, so the local move does contribute mixing once the target is concentrated. Per second that is parity, and at the mild and the density targets the ratio stays at 0.63 and 0.39. No configuration reaches 1.5.

Iowa. The same comparison on Iowa ($k = 4$, $\varepsilon = 0.05$), where there is no ground truth, measures mobility only. We use the paper's three kernels with λ scaled so that $\lambda\, w(\partial P)$ at the initial plan equals the uniform case's $0.1\,|\partial P|$, and run 2×10^5 steps per chain at $p_{\text{loc}} \in \{0.1, 0.3\}$ and both timesteps (Table 7). Real geography reverses the grid's verdict, for the two reasons §4.6 gives. The tolerance makes single flips feasible, so the local move gains its first-order channel and moves 0.13 to 0.17 units per step at the locked timestep, at acceptance 0.2 to 0.7; and a uniform tree edge almost never splits fifty heterogeneous counties within five percent of balance, so the tree move accepts 0.6 to 1.2 percent of proposals. Under the density target the hybrid at the locked timestep reaches 2.9 and 2.1 times the tree move's effective samples per second, and the two halves of its trace agree while the tree move's do not, so within 2×10^5 steps the tree move alone has not reached equilibrium and the hybrid has. Under the perimeter target the short runs are too unstable to quote. The stored session gives ratios of 0.9 and 4.4, an earlier session that matched it for 21 000 steps before diverging through solver sensitivity (Appendix C) gave 1.6 and 3.9, and the tree move's effective sample size is 6. Under the uniform target the evidence is marginal, 1.53 at $p_{\text{loc}} = 0.1$ and 0.38 at 0.3. Raising the timestep to 0.3 does not transfer the grid's optimum. On Iowa the balance window binds more often as τ grows (Figure 2c), so more local solves fall back to the

QP solver, acceptance drops, and the perimeter ratios sit at 1.07 and 1.26 while density holds at 2.8 and 3.4. The tree move's effective sample sizes in every short run are 6 to 14, so the short-run ratios are rough; the direction under the density target and the convergence contrast are not. Runs of 2×10^6 steps settle the rest. Under the perimeter and density targets its quarter means fluctuate around the hybrid's level, 27.6 to 29.4 against 27.9 and 16.4 to 18.3 against 18.2, so the two chains share an equilibrium on real geography, which is the correctness check that no enumeration can give here. Its effective sample size is 28 and 20 over 1.8×10^6 retained steps, an autocorrelation time near 10^5 steps and 0.07 and 0.06 effective samples per second, below the short-run estimates. A matched 2×10^6-step run of the hybrid at $p_{\text{loc}} = 0.1$ gives 126 and 186 effective samples, 0.23 and 0.39 per second, with quarter means on the same level as the tree move's. On equilibrium chains the hybrid's advantage is therefore a factor of 3.1 under the perimeter target and 6.7 under density, larger than the short runs suggested because those understated the tree move's autocorrelation more than the hybrid's. A second seed repeats the picture, with ratios of 3.7 under the perimeter target and 4.7 under density, the hybrid's quarter means within 0.4 of each other, and the tree move's wandering by one to two units over the run.

Reading. The two test beds answer the same question in opposite ways, and §4.6 says why. On an exact-balance grid the local move's leading-order draw is infeasible and its reach is capped, while the tree move splits ten cells evenly often enough to relocate whole districts for microseconds a step; there the local kernel is a passenger. On real geography with a five percent tolerance the local move's first-order channel opens and the tree move's acceptance collapses by an order of magnitude; there the local kernel is what lets the exact sampler reach equilibrium under geography-weighted targets in a run of practical length. The practical exact sampler for a kernel-weighted compactness target is therefore the mixture, with a small local fraction, run at or below the paper's locked timestep rather than at the grid's mobility optimum.

5.4 Iowa: the symmetric-q filter against the exact sampler

On Iowa there is no enumeration, but the exact chains of §4.7 estimate the nominal target directly. Under the uniform kernel at $\lambda = 0.1$ the four hybrid runs of Table 7 put the target's mean cut count near 90, with half-run means between 86 and 95. Tempered ReCom with the symmetric-q filter, run for 500 steps from the same kind of tree-partition start (seed 42, burn-in 100, per-step balance audit passed), sits at 39.8 ± 4.3 cuts, and Vanilla ReCom at 41.4 ± 4.8; Chain E alone, climbing from the same start, averages 55 cuts over steps 100 to 500 and ends the run at 69, still rising. On real geography the chain practitioners run as "targeting π" therefore samples plans with less than half the boundary length that π itself has. The 5×5 calibration showed the effect at the level of distributions; Iowa shows its size.

6 Discussion

6.1 What each chain is for

Chain E is provably exact for its target and, alone, too immobile to be a sampler; its value is as the local component of the hybrid, and §5.3 shows that value is nil on an exact-balance grid and decisive on Iowa under weighted targets. **The exact tree move** is the mobile component, exact for any kernel-weighted target once Metropolized to it; its cost is acceptance. **The hybrid** is the practical sampler, with a small local fraction and a timestep at or below the locked one. **Chains A and V** are what practitioners run, and §5.1 and §5.4 measure how far their output sits from the tempered target.

Why does the fidelity term matter? Without it the local move is a fresh spectral re-split that discards the structure the chain has found; with it the move is anchored and its size is governed by Corollary 4.1. The companion paper shows on Iowa that only the anchored integer form of the energy improves compact-

Target	Chain	acc tree	acc loc	linear	$\bar{w}$ 1st half	$\bar{w}$ 2nd half	ESS	ESS/s	ratio
locked timestep $\tau = 0.1$									
uniform, λ=0.10	tree only	0.012	—	—	83.2	87.2	14	0.38	—
	hybrid p=0.1	0.005	0.265	0.58	89.3	86.8	37	0.59	1.53
	hybrid p=0.3	0.004	0.214	0.51	88.4	85.7	18	0.15	0.38
perimeter, λ=0.34	tree only	0.006	—	—	25.5	28.2	6	0.16	—
	hybrid p=0.1	0.005	0.560	0.81	26.6	29.0	8	0.15	0.90
	hybrid p=0.3	0.005	0.532	0.80	28.1	27.7	67	0.72	4.44
density, λ=0.72	tree only	0.007	—	—	13.3	14.8	6	0.20	—
	hybrid p=0.1	0.004	0.615	0.97	18.8	18.2	28	0.58	**2.92**
	hybrid p=0.3	0.004	0.676	0.98	18.9	18.1	28	0.43	2.14
grid mobility optimum $\tau = 0.3$									
uniform	hybrid p=0.1	0.003	0.097	0.38	90.7	90.4	39	0.57	1.50
	hybrid p=0.3	0.006	0.117	0.40	95.5	90.8	18	0.14	0.37
perimeter	hybrid p=0.1	0.007	0.249	0.52	27.7	29.1	12	0.17	1.07
	hybrid p=0.3	0.006	0.274	0.59	28.8	28.4	24	0.20	1.26
density	hybrid p=0.1	0.005	0.449	0.95	17.0	17.5	27	0.57	2.84
	hybrid p=0.3	0.005	0.396	0.95	17.6	18.2	47	0.69	**3.44**
long runs, 2×10^6 *steps,* $\tau = 0.1$, *seed 42*									
perimeter	tree only	0.003	—	—	28.2	28.5	28	0.073	—
	hybrid p=0.1	0.005	0.539	—	28.0	28.7	126	0.226	**3.1**
density	tree only	0.005	—	—	17.3	17.4	20	0.059	—
	hybrid p=0.1	0.005	0.672	—	18.0	18.0	186	0.394	**6.7**
long runs, 2×10^6 *steps,* $\tau = 0.1$, *seed 43*									
perimeter	tree only	0.004	—	—	28.7	29.1	26	0.068	—
	hybrid p=0.1	0.005	0.527	—	28.4	28.3	147	0.253	**3.7**
density	tree only	0.004	—	—	16.6	17.6	39	0.108	—
	hybrid p=0.1	0.005	0.678	—	18.2	18.2	243	0.507	**4.7**

Table 7: The exact tree move and the hybrid on Iowa ($k = 4$, $\varepsilon = 0.05$), seed 42 except where marked, 2×10^5 steps each except the last two blocks. "linear" is the fraction of local solves that the linear-solve path of Lemma 4.1 handled without the QP solver; the two $\bar{w}$ columns are the mean weighted cut over the first and second halves of the retained trace, whose agreement is the convergence check; "ratio" is effective samples per second relative to the tree move alone at the same target, timed in the same session, so the $\tau = 0.3$ ratios use that session's tree-only wall time rather than the displayed row's.

ness over its chain family, while an unanchored spectral variant matches plain ReCom. The community step-trace diagnostic (Appendix B) makes a complementary point about locality in general. Pure tree-move chains cross community boundaries at 13–15% per step, near the community-blind baseline, while optimization-guided local mixtures drop to 5–11%, which is the division of labor the hybrid formalizes.

6.2 On the macro-structure findings

The corrected picture of Appendix B has two consequences. First, it relocates the blame for slow local mixing. If plan space were islands joined by narrow corridors, slow traversal would be a topological inevitability; since the fibers are well connected, it is a property of the dynamics, and §4.6 identifies that property for Chain E as the feasibility of its draws, not the graph. Second, worst-case hardness results [2, 15] permit bottlenecked instances and Gold [30] exhibits planar-grid configurations that trap recombination for exponentially many expected steps, so our grids may simply be too small or too symmetric to show such structure; the statistics of Table 10 give the template for checking a fiber that does.

6.3 Limitations

(L1) No theoretical mixing-time bound. Like every redistricting chain used in practice, we lack a polynomial mixing-time guarantee (the polynomial-time grid sampler of [13] is not a local chain), and Chain E demonstrates that even an exactly reversible chain can be an extremely slow mixer. We rely on empirical diagnostics, namely ESS, the community step-trace diagnostic, and enumerated ground truth.

(L2) Chain A's symmetric-q approximation is not a small bias. On the enumerated fiber the tempered chain's empirical mean (16.73) sits 1.8 cuts below the exact target mean (18.56), its cut distribution is 220 noise-floor standard deviations from that target (§5.1), and Chain E's Iowa trajectory suggests the gap is larger, not smaller, on real graphs. §4.7 corrects it with the spanning-tree-count ratio, at the price of acceptance.

(L3) Chain E mixes too slowly to use alone. Exactness comes from single-shot rounding, whose reach Corollary 4.1 bounds and whose feasibility §4.6 measures; the chain visited 6 of 4 006 fiber states in 1 000 steps at the locked timestep, which §4.6 shows sits well below the mobility optimum. Alone it is not a sampler; in the hybrid of §4.7 it is a component whose value depends on the geography, nil on an exact-balance grid and decisive on Iowa under weighted targets.

(L4) Several runs are single-seed. The baseline tables and the hybrid runs report seed 42; the calibration quotes ten-seed means of Chains V and A, the hybrid's long-run Iowa ratios rest on two seeds, and its short-run ratios on one.

6.4 Future directions

1. **A more mobile exact tree move.** The hybrid of §4.7 is exact, but its tree component accepts under two percent of proposals on real geography because it draws the cut edge among all tree edges. Recovering GerryChain's balanced-edge selection without losing the closed form means lifting the state to a forest, as Metropolized forest recombination does [4]; combining that lift with the matched local kernel is the direct next step, and the criterion of §5.3 is the test to apply to it. The second step is precinct scale, where the determinants are larger but still cheap, and where the downstream question, whether the choice of kernel changes an ensemble's partisan statistics, can be asked on real elections.

2. Integration with the cycle walk [24] or the balanced up-down walk [1] as the global component in place of the tree move.

3. Multi-scale diffusion kernels (heat kernel, commute-time kernel) in the local move.

7 Conclusion

This paper introduced a diffusion proposal for redistricting Markov chains, one Laplacian energy $\alpha\, x^\top L x + \beta\, \|x - x_0\|^2$ under global population balance, and showed that the diffusion view is more than an interpretation. The proposal is a single linear solve whenever balance is slack, its expected move size is an explicit filter norm with a small-timestep law and a hard ceiling, and its mobility is governed by feasibility rather than by the Metropolis ratio, with an interior optimum in the timestep that the locked setting missed by a factor of about four. Corrected by an exact Metropolis–Hastings step, the proposal gives Chain E, to our knowledge the first optimization-based proposal in redistricting MCMC with a detailed-balance argument, and on an 18-state fiber a million-step run's occupancy distribution is not distinguishable from the exact target. Paired with an exact tree move it gives a reversible sampler for kernel-weighted compactness targets, validated on 4 006 enumerated states, in which the local kernel is a passenger on an exact-balance grid and the component that reaches equilibrium on Iowa under perimeter and density targets. On the enumerated 5×5 fiber we showed at the level of distributions that tempered ReCom does not sample its nominal target and sits close to the spanning-tree-weighted measure, while Chain E samples what it states. The same energy solved as an integer program is an anchored refinement tool whose evaluation as an optimizer is the subject of a companion paper. Per-step balance and ESS audits discipline the empirical claims, including our own.

For the field, we hope two ideas outlast the specific chains. Kernel choice can be a first-class modeling decision inside the proposal and the target rather than a post-hoc filter, provided the induced distribution is corrected. And enumerated fibers are cheap, severe test beds; a chain's distance from its own computable target on 4 006 states, read against a noise floor, is a more informative diagnostic than any p-value on a single trace.

Code and data availability

All code, the enumerated fibers, and every artifact behind every table and figure are in the public repository `https://github.com/kirtisoglu/Redistricting_Diffusion_Model`, under `diffusion_model/recom/`, with the scripts and command lines listed in Appendix C. An archived release with a persistent identifier will be deposited on acceptance.

A Algorithm details and hyperparameters

A.1 Common setup

- Dual graph $G = (V, E)$ with vertex populations p_v and edge attributes (`shared_perim` for Iowa).
- k districts, ε-population balance per Equation (2), enforced as a two-sided bound around the global ideal on every move (Equation (4)) and audited per step. The bare GerryChain `recom` proposal is wrapped with an explicit whole-plan balance check (`recom_valid`), since its internal cut search validates only one side of each split.
- Initial partition from `gerrychain.tree.recursive_tree_part`, redrawn until all k districts satisfy the global tolerance (`balanced_tree_part`).

- Step count: 5×5: $N = 1\,000$, burn-in $B = 100$; Iowa: $N = 500$, $B = 100$; long-run validation (§5.1): 2×10^5 (A) and 10^5 (E) steps; hybrid and tree-move runs (§5.3): 2×10^5, and 2×10^6 for the matched Iowa runs.
- Random seeds: base seed 42 with a distinct offset per chain (independent streams for Python `random` and `numpy`).

A.2 Chain-specific parameters

Chain	Parameter	Value
A	λ (target temperature)	0.10
E	α	1.0
E	β	10.0 ($\tau = \alpha/\beta = 0.1$; Corollary 4.1 gives the resulting move size and §4.6 places the mobility
E	Rounding	single-shot Bernoulli($x^\star$)
E	λ (MH target)	0.10
T	cut edge	uniform over all tree edges; fair-coin labels
H	p_{loc}	0.1 (Iowa), 0.2 and 0.5 (5×5)
H	$\tau = \alpha/\beta$	0.1, and 0.3 where stated

Table 8: Locked-in hyperparameters for the chains of Table 1.

A.3 Compute environment

- macOS (Darwin 23.6.0), Python 3.12.
- Gurobi 13.0.2 (academic license) for the QP solves that the linear-solve path of Lemma 4.1 does not cover, at its default thread count.
- GerryChain 0.2.22 for the ReCom proposal and the initial tree partitions; NetworkX 3.x for graph operations.
- One Python process per run; no GPU. All experiments were run on a 2024 MacBook.

B Partition fiber structure: communities without bottlenecks

This appendix documents a structural diagnostic of the partition fiber $\mathcal{P}_k(G;\varepsilon)$ (the set of all valid plans), viewed as a graph under its minimal valid move relation. Two findings emerge, one positive and one negative, and we report both. Positively, all three enumerated fibers carry genuine mesoscale community structure, with modularity far above degree-preserving null models, and chains differ sharply in how often they cross community boundaries. Negatively, a plausible folk intuition (that the communities are "islands" joined by narrow corridors or bottleneck edges) is refuted by method-independent statistics. Community interfaces are wide, nearly all community pairs are directly connected, and the literal bridges of the exact-balance fiber turn out to be pendant vertices. The diagnostic still motivates the local-refinement design of §4, and we believe both findings are of independent interest for analyzing redistricting MCMC.

Fiber	Nodes	Edges	k	ε	Sizes	Adjacency
5×5, $\varepsilon = 0$	4 006	13 416	5	0	{5}	boundary swap
5×5, $\varepsilon = 0.20$	193 128	969 760	5	0.20	{4, 5, 6}	single flip
4×4, $\varepsilon = 0.25$	1 953	7 024	4	0.25	{3, 4, 5}	single flip

Table 9: The three test-bed fibers. Adjacency is the *minimal valid move* of each fiber. At $\varepsilon = 0$ no single-vertex reassignment preserves exact balance (it produces districts of sizes 4 and 6), so adjacency is the boundary *swap*, which exchanges one boundary vertex of district a with one of district b, both districts remaining connected. At $\varepsilon > 0$ adjacency is the single *flip*, which reassigns one boundary vertex, both affected districts remaining connected and within size bounds. Cross-fiber comparisons carry one caveat. The $\varepsilon = 0$ fiber lives under a different (two-vertex) move relation than the $\varepsilon > 0$ fibers.

Three test beds. We construct three fully enumerated partition fibers spanning grid size, district count, and balance tolerance.

For each fiber, all valid partitions are enumerated and the single-flip adjacency graph is built explicitly (enumeration runtimes on a 2024 MacBook range from ~ 5 s for the 4×4, $\varepsilon = 0.25$ fiber to ~ 50 min for the 5×5, $\varepsilon = 0.20$ fiber; scripts accompany the replication materials).

Community structure, validated against null models. Each fiber graph carries substantial mesoscale community structure under modularity maximization. We report the method per fiber (greedy modularity, Clauset–Newman–Moore, for the $\varepsilon = 0$ and 4×4 fibers; Louvain, resolution 1.0, seed 42, for the large $\varepsilon = 0.20$ fiber), validate against degree-preserving rewiring nulls, and treat the specific community counts as method-dependent descriptive statistics.

	Community structure				Anti-bottleneck statistics		
Fiber	Comm.	Q	Null Q	Method	Pairs conn.	Inter-comm. edges	Width (med/max)
5×5, $\varepsilon = 0$	10	0.65	0.36	CNM	15/15	18.3%	142 / 383
5×5, $\varepsilon = 0.20$	26	0.69	n/a	Louvain	320/325	26.8%	667 / 3 006
4×4, $\varepsilon = 0.25$	7	0.58	0.34	CNM	21/21	23.3%	71 / 186

Table 10: Left: community decompositions with degree-preserving null baselines (mean Q of CNM on double-edge-swap rewirings of the same graph; the 193k-node fiber's null is omitted for cost). Real Q exceeds null Q by 0.24–0.29, so the community structure is genuine, not a degree-sequence artifact. Counts vary with the detection method and should be read qualitatively. Right: the statistics that refute the corridor picture. Nearly every pair of communities shares a direct interface ("pairs conn." counts the large-community pairs at $\varepsilon = 0$), interfaces are tens to thousands of edges wide, and a fifth to a quarter of all fiber edges run between communities. The exact-balance fiber's 16 literal single-edge bridges each split off a *single vertex* (pendant leaves), not an island, and the six large $\varepsilon = 0$ communities have conductance 0.15–0.22, an order of magnitude above bottleneck territory.

No narrow corridors: a corrected intuition. It is tempting to picture these communities as islands joined by a few narrow corridors, with bottleneck edges controlling passage. The method-independent statistics on the right of Table 10 refute that picture on all three fibers.

Concretely, on the $\varepsilon = 0.20$ fiber, interfaces thinner than 50 edges do exist (there are 16 of them), but they are simply the tail of a width distribution whose median is 667 and whose top interfaces carry 3 000 edges; a chain moving through this graph crosses between communities through the wide interfaces,

and the thin tail is irrelevant to mixing. The correct summary of fiber macro-structure is therefore *communities without bottlenecks*. Valid-plan space is regionally organized but well connected, and slow traversal by local chains (below) is a property of the *dynamics*, not of corridor topology. This negative result matters for chain design. It says local chains stall not because the state space is nearly disconnected but because their moves are small and, in Chain E's case, because distant proposals are rarely feasible (§4.6); global recombination moves, by contrast, relocate across communities in a single step.

Chain step-trace diagnostic. The community decomposition supports a direct diagnostic for any redistricting chain. We label each visited partition by its greedy-modularity community on the $\varepsilon = 0$ fiber and record the *jump rate*, the fraction of steps that move to a different community.

Chain	Communities visited	Jump rate	Jumps in 1 000 steps
V: Vanilla ReCom	8/10	15.3%	153
A: Tempered ReCom	7/10	12.7%	127
B: QP-diffusion (50/50)	6/10	4.6%	46
C: ReCom + IQP local	6/10	10.5%	105
D: ReCom + spectral mix	7/10	9.3%	93
E: QP-MH exact	2/10	0.3%	3

Table 11: Inter-community jump rates on the 5×5, $\varepsilon = 0$ fiber (1 000 steps per chain, seed 42, per-chain RNG streams). A community-blind baseline crosses at roughly the inter-community edge share (18.3%, Table 10); ReCom variants (V, A) sit near it at $\sim$ 13–15% per step, the local kernels (B, C, D) are stickier at $\sim$ 5–11%, and the strictly local QP-MH chain (E) almost never crosses, a property of its small accepted moves rather than of any bottleneck in the fiber. Chains B, C and D are the companion paper's continuous, integer and spectral mixtures, included here because the diagnostic was run on all six.

Reading of the diagnostic. ReCom relocates across communities in single steps; optimization-guided local kernels work within them; and the exact-but-strictly-local Chain E shows the failure mode of small accepted moves. (Per-chain step-traces, which show the same contrast as dwell patterns over time, are released as supplementary figures.) The regimes are complementary, and they suggest an explicit design principle. A good redistricting sampler should combine global moves that relocate across regions of plan space with local moves that refine within them; ReCom does the former, anchored local kernels the latter. This is the design of the hybrid of §4.7 and of the companion paper's Chains B and C, each step a random mixture of a tree move and a diffusion move. We note the diagnostic's limits (one run per chain on one fiber, and community labels from a specific detection method) and read it as motivating evidence, not proof.

C Reproducibility checklist

- Code: `diffusion_model/recom/` in the project repository.
- Modules: `problems.py` (graph setup, balanced initial plans), `chains.py` (chain runners, balance-validated ReCom wrapper, per-step audit), `qp_model.py` (QP/IQP/spectral/QP-MH step factories, global-ideal balance windows), `diagnostics.py` (KL/TV diagnostics).
- Orchestrators: `iowa_target_matched.py` (the baseline chains V, A and E; mode, seed and kernel as CLI arguments), `long_run_5x5.py` and `matrix_tree_targets.py` (§5.1), `calibration_distance.py` and `fig_calibration.py` (§5.1, reading only stored artifacts), `db_check_2x3.py` (Remark 4.1), `long_run_2x3.py` with `gof_2x3.py` (§5.2; arguments `1000000 42`), `spectral_move_size.py` and

`fig_spectral.py` (§4.3), `acceptance_sweep.py` (Table 2), `mobility_law.py` (Table 3), `hybrid.py` (the exact tree move, the hybrid, and the analytic kernel check of §4.7) with `hybrid_validate_5x5.py` (Table 6; arguments `200000 42 0.5 0.1` and `200000 42 0.2 0.3`), `hybrid_iowa.py` (Table 7; arguments `200000 42 0.1` and `200000 42 0.3`) with `hybrid_iowa_long.py` (the 2×10^6-step rows; arguments `tree 2000000` and `hybrid 2000000`, with `42` or `43` as a third argument for the seed), and `ten_seed_stats.py` (between-seed statistics).

- Outputs, all under `iowa_hybrid_test_out/`: `target_matched_*_seed*.npz` for the baseline chains, each stamped with the full configuration and containing per-step cut traces, timings, and balance-audit arrays; `long_run_5x5_seed42.npz`, `matrix_tree_targets.npz` and `calibration_distance_seed42.npz` for §5.1; `long_run_2x3_n*_seed42.npz` for §5.2; `spectral_move_size.npz`, `acceptance_sweep_5x5.npz` and `mobility_law_5x5.npz` for §4.3 and §4.6; and `hybrid_*.npz` for §5.3, so every table can be regenerated from its own artifact.

- Fibers: `fiber_5x5/` contains the 5×5 and 4×4 enumeration scripts and precomputed adjacency files.

- Random seeds: base 42, per-chain offsets. Gurobi default seed (deterministic given the same input).

- Reproducibility across sessions. The 5×5 runs and the Iowa runs under the uniform and density kernels reproduce bit for bit from their seeds. The Iowa runs under the perimeter kernel do not: a repeat of the $\tau = 0.1$ session matched the stored trajectories for 21 217 and 185 390 steps and then diverged, which we attribute to floating-point sensitivity of the QP solver fallback under that kernel's wide range of edge weights. The stored artifacts are the realizations reported, the earlier session is kept as `hybrid_iowa_tau0.1_earlier_session.npz`, and the perimeter-target conclusions rest on the long runs rather than on any single short run.

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